

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science and Engineering
MID SEMESTER EXAMINATION, Jan-June 2023

B Tech: Semester-IV

Course Name: Computer Networks

Course Code: CS44113

Maximum Time: 2 hours

Max. Marks: 30

Instructions:

1. Attempt all questions.
2. Assume any suitable data, if necessary. *(Any other instruction need to provide by the concern faculty)*
3. The marks, Course Outcome (CO) and Bloom's Level (BL) related to questions are mentioned on the right-hand side margin.

		Marks	CO	BL
1	Examine and discuss at least five recent applications of computer networks. How would your life be altered if these networks were suddenly switched off?	6	CO1	Creating and Understanding
2	Explain the difference between OSI and the TCP/IP reference model in brief with suitable diagram. Further, discuss the major functionality and important services of TCP/IP reference model.	6	CO2	Evaluating and Creating
3	Compare and contrast multiplexing techniques is (are) used to combine analog and digital signals? How is the wastage of bandwidth in synchronous TDM overcome by Asynchronous TDM? A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?	6	CO2	Analyzing and Creating
4	List the difference (at least five) between circuit switching, datagram packet switching and virtual-circuit packet switching. Suppose that 'M' bits of user data are to be transmitted over a k-hop path in a packet-switched network as a series of packets, each containing 'X' data bits and 'P' header bits, with $M > X + P$. The bit rate of the lines is 'B' bps and the propagation delay is negligible. Determine the end-to-end delay and what value of 'X' minimizes the total delay?	6	CO2	Remembering and Evaluating
5	Explain the potential difference between error detection codes with suitable diagram? A bit stream 110101010 is transmitted using the standard CRC method. The generator polynomial is $X^4 + X^2 + 1$. Show the actual bit string transmitted. Suppose the 4 th , 5 th and 6 th bit from the left is inverted during transmission. Show that this error is detected at the receiver's end.	6	CO3	Evaluating and Remembering